

ACTIVITY 14 — HACKATHON PREPARATION

1. Activity Details

Activity Number: 14

Activity Title: Hackathon Preparation

Hackathon: InnoHacks 4.0

Organizer: Innogeeks

Venue: KIET Deemed to be University, Ghaziabad

Platform: Devfolio

Team Size: 4 Members

Selected Track: Open Innovation

Project Name: TeamMatch — Student Team Formation Platform

Official Hackathon Sources

Official InnoHacks 4.0 Website:

<https://www.innohacks.live/>

Official InnoHacks 4.0 Devfolio Page:

<https://innohacks-4.devfolio.co/>

The official InnoHacks 4.0 website lists four tracks:

- IoT
- Agentic AI/GenAI
- Blockchain/Web3
- Open Innovation

The team selected the **Open Innovation** track because it allows participants to develop a solution for a real-world problem and demonstrate the solution within the hackathon.

The official Devfolio page describes the first stage as the **Idea PPT Submission Round**, requiring teams to submit an idea containing the problem statement, proposed solution, technology stack, and implementation approach.

Official Timeline

According to the official InnoHacks 4.0 website:

- **Registrations Open:** September 1, 2026
- **PPT Submissions Begin:** September 2, 2026
- **PPT Submissions Close:** September 30, 2026

- **Team Shortlisting Ends:** October 3, 2026
- **Hackathon Begins:** October 10, 2026
- **Hackathon Ends:** October 11, 2026

The official Devfolio page describes the offline hackathon as a 24-hour event for shortlisted teams.

2. Team Members

Sl. No.	Name	Branch / Semester	Role
1	T S Harshavardhan Nayaka	CSIT – 3rd Sem	Frontend
2	Vighnesh P K	CSIT – 3rd Sem	Backend
3	Terrance Paul	CSIT – 3rd Sem	All-rounder
4	SHASHANK G NAIK	CSIT – 3rd Sem	UI/UX

3. Team Responsibilities

T S Harshavardhan Nayaka — Frontend

Planned Contribution:

Responsible for planning and developing the user-facing components of the TeamMatch platform, including webpages, navigation, project discovery interfaces, and student-facing components.

Vighnesh P K — Backend

Planned Contribution:

Responsible for backend logic, application flow, server-side functionality, and supporting communication between the frontend and database.

Terrance Paul — All-rounder

Planned Contribution:

Responsible for supporting different areas of the project, including technical research, integration, testing, development support, and coordination between team members.

SHASHANK G NAIK — UI/UX

Planned Contribution:

Responsible for planning the user interface structure and user flow, focusing on clear navigation, simple student profiles, project discovery, and an easy team-matching experience.

4. Hackathon Track Selection

The team selected the **Open Innovation** track of InnoHacks 4.0.

The official InnoHacks website describes Open Innovation as a track without a fixed technical restriction, where participants can build a solution to a real problem and demonstrate it within the available hackathon period.

Selected Track

Open Innovation

Reason for Selecting the Track

The team wanted to work on a student-focused problem that could be addressed through a web-based technology solution.

The selected concept focuses on helping students find suitable teammates for:

- Academic projects
 - Hackathons
 - Technical competitions
 - Innovation projects
 - Collaborative development
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5. Proposed Problem Statement

TeamMatch — Student Team Formation Platform

Students frequently need teammates for academic projects, hackathons, technical competitions, and collaborative activities.

However, students may not know which other students have the required technical skills, interests, preferred roles, or availability for a particular project.

Finding suitable teammates through informal communication, personal contacts, or separate messaging groups can therefore be difficult.

Problem Statement

How can students be provided with a structured platform to discover suitable teammates based on their skills, interests, preferred roles, and project requirements?

6. Proposed Solution

TeamMatch

TeamMatch is a proposed student-focused platform that helps users discover and connect with suitable teammates based on their:

- Technical skills
- Interests
- Preferred roles
- Project requirements
- Experience
- Availability

The platform would allow students to create profiles, create or discover projects, identify required roles, and connect with potential teammates.

The concept is intended to provide a structured approach to student team formation.

7. Main Objectives

The proposed platform aims to:

1. Help students find suitable teammates.
2. Make student skills easier to discover.
3. Allow students to specify their preferred project roles.
4. Help project creators identify missing team roles.

5. Provide a structured way to search for potential teammates.
 6. Make collaboration easier for hackathons and academic projects.
 7. Help students discover projects matching their interests and skills.
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8. Proposed Features

8.1 Student Registration

Students can create an account and provide information such as:

- Name
 - Branch
 - Semester
 - Skills
 - Interests
 - Preferred role
 - Experience level
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8.2 Student Profile

Each student would have a profile containing relevant information.

Example

Name: Student Name

Branch: CSIT

Semester: 3rd

Skills: C, Python, HTML, CSS

Preferred Role: Frontend Developer

Interests: Web Development, AI, Hackathons

8.3 Project Creation

A student can create a project and provide:

- Project name
- Project description
- Required skills
- Required team roles

- Number of members required
 - Project category
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8.4 Project Discovery

Students can browse available projects and identify projects that match their interests and skills.

8.5 Team Matching

The proposed system can compare:

Student Skills ↔ Required Project Skills

and identify students whose skills and preferred roles match project requirements.

A simple rule-based matching approach can initially be used.

8.6 Team Requests

Students can send requests to join a project or invite another student to their team.

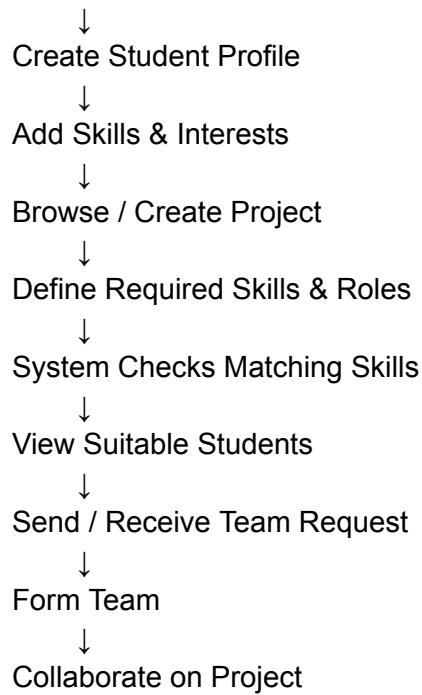
8.7 Dashboard

A student dashboard can display:

- Profile information
 - Created projects
 - Joined projects
 - Team requests
 - Matching projects
 - Potential teammates
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9. Proposed User Flow

Student Registration



10. Proposed Matching Logic

The initial concept uses a simple rule-based matching mechanism.

Example

Required Project Skills:

Python
HTML
CSS
Database

Student Skills:

Python
HTML
CSS



3 out of 4 required skills match



Student identified as a potential match

The system can consider:

- Skill overlap
- Preferred role
- Project requirements
- Student interests

A matching score could be introduced in a future version.

11. Proposed Technology Stack

Frontend

- HTML
- CSS
- JavaScript

A framework such as React could be considered for an advanced implementation.

Backend

- Python
- Flask

Database

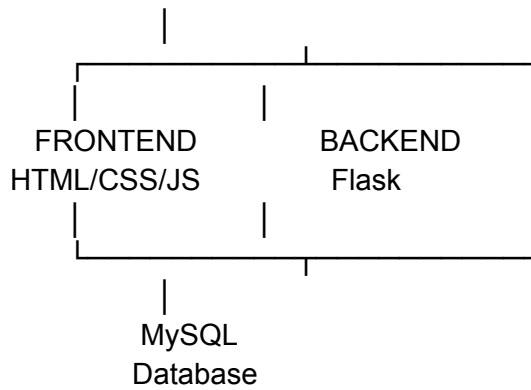
- MySQL

Development Tools

- Visual Studio Code
 - Git
 - GitHub
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12. Proposed System Architecture

TEAMMATCH



13. Functional Modules

Module 1 — Student Registration

Handles student account creation and basic information.

Module 2 — Student Profile

Stores skills, interests, preferred roles, and other relevant information.

Module 3 — Project Management

Allows students to create projects and specify required skills and roles.

Module 4 — Search & Discovery

Allows students to search for projects and potential teammates.

Module 5 — Matching

Compares student skills and preferences with project requirements.

Module 6 — Team Requests

Handles invitations and requests between students.

Module 7 — Dashboard

Provides an overview of projects, requests, and possible matches.

14. Example Use Case

Scenario

A student wants to participate in a hackathon and needs:

- 1 Frontend Developer
- 1 Backend Developer
- 1 UI/UX Designer

The student creates a project on TeamMatch and specifies these requirements.

Other students create profiles containing their skills and preferred roles.

The system compares project requirements with available student profiles.

Potential matches are displayed to the project creator.

The students can then send requests and form a team.

15. UI/UX Planning

The UI/UX design will focus on keeping the platform simple and easy to navigate.

Proposed Main Screens

1. Login / Registration
2. Student Profile
3. Home Dashboard
4. Explore Projects
5. Create Project
6. Project Details
7. Find Teammates
8. Team Requests
9. My Teams

Design Principles

- Simple navigation
- Clear buttons and labels
- Easy-to-read student profiles
- Minimal unnecessary information

- Consistent layout
 - Mobile-friendly design where possible
 - Clear distinction between projects and student profiles
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16. Innovation / Uniqueness

The proposed idea focuses specifically on **student team formation**.

Instead of students depending only on personal contacts or informal college groups, TeamMatch proposes a structured platform where students can:

- Display their skills
- Find relevant projects
- Identify missing team roles
- Discover students with complementary skills
- Send team requests

The matching concept provides a structured approach to team formation.

17. Expected Benefits

The proposed platform could help:

- Students find teammates more easily.
 - Students discover projects matching their interests.
 - Project creators identify missing skills.
 - Students showcase their technical abilities.
 - Hackathon teams form more efficiently.
 - Students collaborate outside their immediate friend groups.
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18. Future Enhancements

Possible future improvements include:

- Advanced skill-based matching
- Profile verification
- GitHub profile integration
- LinkedIn profile integration

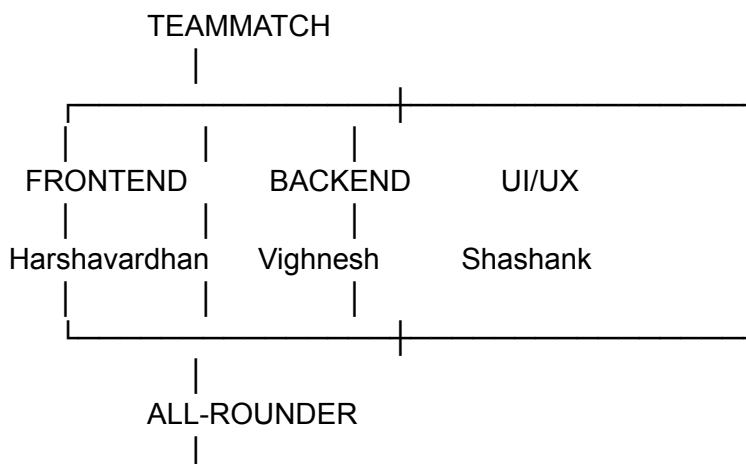
- Project ratings
 - Team reputation
 - Chat functionality
 - Notifications
 - Advanced search filters
 - AI-assisted teammate recommendations
 - Skill-gap identification
 - Project recommendation system
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19. Hackathon Preparation Work

As part of the activity, the team focused on:

- Understanding the hackathon format.
 - Researching possible problem areas.
 - Reviewing the available hackathon tracks.
 - Selecting a suitable track.
 - Discussing potential project ideas.
 - Identifying the target users.
 - Defining the proposed problem statement.
 - Developing the TeamMatch solution concept.
 - Identifying major system features.
 - Dividing responsibilities among team members.
 - Discussing the proposed technology stack.
 - Planning the basic system architecture.
 - Preparing the initial project documentation.
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20. Team Role Distribution



21. Repository

The preparation work and related documentation are maintained in the project repository.

GitHub Repository:

<https://github.com/harshavardhannayaka20072007-png/Activity-14-hackathon-preparation>

22. Official References

InnoHacks 4.0 Official Website

<https://www.innohacks.live/>

The official website provides information about the hackathon, tracks, timeline, venue, participation, and event structure.

InnoHacks 4.0 — Devfolio

<https://innohacks-4.devfolio.co/>

The Devfolio page provides the official hackathon information and Idea PPT submission details.

23. Activity Outcome

Through this activity, the team developed an initial understanding of hackathon preparation, problem identification, solution planning, team role distribution, feature identification, and basic technical architecture.

The team selected the **Open Innovation** track and developed **TeamMatch** as a proposed solution concept for helping students form teams based on skills, interests, preferred roles, and project requirements.

The activity resulted in a structured project concept containing:

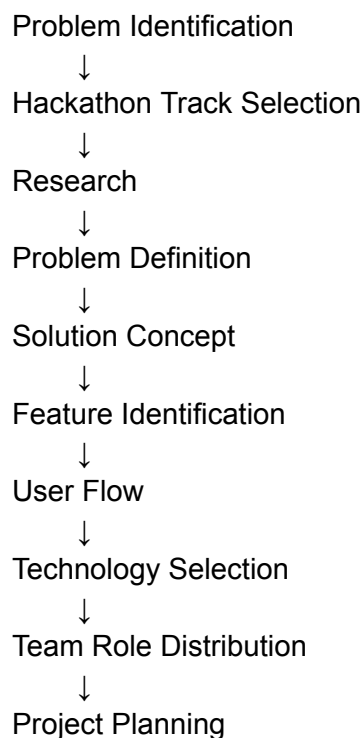
- Problem statement
 - Target users
 - Proposed solution
 - Features
 - User flow
 - Matching approach
 - Technology stack
 - System architecture
 - Team responsibilities
 - Future enhancements
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24. Conclusion

The activity provided practical exposure to the early stages of hackathon preparation and project planning.

The team identified a student collaboration problem and proposed **TeamMatch**, a platform concept for structured student team formation.

The activity demonstrated the progression:



This preparation provides a foundation for future implementation and hackathon development.

Documentation Note

This document records the team's **hackathon preparation, research, planning, and proposed solution concept**. It does not claim that a complete production system was developed, deployed, shortlisted, or won at InnoHacks 4.0.